
PROFESSIONAL SUMMARY

Data Engineer with ~4 years' experience building and supporting secure, reliable data pipelines, data warehouses/lakes, and analytics-ready models. Strong Python and SQL with proven impact: delivered metadata-driven pipelines processing ~1M+ rows/week, cut daily refresh time ~50%, and improved query performance ~35% via partitioning/indexing. Hands-on with modern lakehouse patterns (Databricks Autoloader, DLT, Unity Catalog) and familiar with AWS ecosystem through project work including S3 and warehouse/lake concepts. Comfortable partnering with analysts/data scientists and supporting production for global stakeholders

EXPERIENCE

Data Engineer**May 2024 – Present****The DRG, United Kingdom**

- Designed and optimized scalable **data models** supporting Finance and Ops; improved **data integrity**, accessibility, and **governance** aligned to enterprise reporting and control requirements
- **Built and maintained** parameterized data pipelines processing **~1M+ rows/week** from EPOS, loyalty, and REST APIs; standardized schemas with **idempotent loads** and drift-tolerant ingestion.
- Optimized orchestration and compute paths, reducing daily refresh **~50%** and improving **SLA** reliability for **Finance/Ops** reporting; strengthened monitoring/alerting for production support.
- Led MySQL to Azure **PostgreSQL** migration and **dimensional re-model**; applied targeted **indexing and partitioning** to reduce key report query times **~35%**, improving warehouse performance and user experience.
- Developed **Python** automation for daily ingestion from **REST APIs** and **semi-structured sources (JSON/XML)**, improved reliability and downstream usability through standardized transformations and validation checks.
- Delivered governed 'gold' datasets with **RLS** and **incremental strategies**; improved data trust and reduced refresh failures through better **lineage**, quality checks, and **operational monitoring**."
- **Partnered with Marketing/Ops** to translate analytical requirements into **engineered datasets** and **measurable experiments**; enabled faster, data-driven planning.
- Enabled near **real-time operational reporting** by improving upstream pipeline **reliability and performance**; supported measurable business outcomes through trusted, timely data.

Data Analyst**April 2021 - March 2022****NSEIT, Nashik**

- **Automated** SQL/Python ingestion and preprocessing with parameterized validations (types, duplicates, referential checks), **freeing ~10 hrs/week** and shortening release cycles.
- **Co-designed** a governed preprocessing framework with a **6-person** team (standardized schemas, data dictionary, access controls), **reducing rework** and handoffs across monthly releases.
- Automated recurring reconciliations and data-processing workflows, freeing **~10 hours/week** and shortening analysis turnaround for stakeholders.
- Generated comprehensive 20+ reports and dashboards to support informed decision-making by stakeholders.
- Monitored data integrity and quality, promptly addressing discrepancies and improving reporting reliability.

Data Analyst**March 2020 – March 2021****Swami Vivekanand Education and Research Centre, Mumbai**

- Integrated and standardized multi-source data with Python (Pandas/NumPy), cutting processing time by **~20%** and improving data reliability for downstream reporting.
- Built interactive **Power BI** dashboards for KPI tracking, enabling self-service analysis and faster decision cycles.
- Authored and optimized complex **SQL** to profile and segment customers, delivering a **~25%** uplift in decision-ready insights for stakeholders.
- Increased stakeholder engagement by **~30%** through targeted walkthroughs, clear documentation, and iterative dashboard improvements.

SKILLS

- **Cloud/Platform:** AWS (S3, Glue, Athena, Redshift, VPC), Azure (ADLS Gen2, Azure SQL Database, Key Vault), Databricks (DLT, Autoloader, Structured Streaming, Unity Catalog, SQL Warehouse), Snowflake, Kafka
- **Data Engineering:** ETL/ELT, CDC, SCD Types, Dimensional Modeling (Star/Snowflake), data warehouse & data lake/lakehouse, Medallion Architecture, Data Quality (DLT Expectations), Lineage/Observability
- **Orchestration/Automation:** Azure Data Factory (pipelines, Mapping Data Flows, SHIR), Azure Logic Apps, GitHub Actions, Databricks Asset Bundles (CI/CD), Airflow
- **Programming/Query:** Python (Pandas, NumPy, PySpark, SQLAlchemy, Flask), SQL/T-SQL (CTEs, window functions, performance tuning), Bash, DAX, PostgreSQL, MySQL, REST/HTTP APIs, JSON/XML

- **Analytics/BI:** Power BI (DAX, RLS, Incremental Refresh, semantic models), Excel (Pivot Tables, VBA Automation), Grafana, Matplotlib, Seaborn
- **Data Acquisition:** REST APIs, BeautifulSoup, Selenium, Web Scraping, Managed Identity Authentication, Python Automation Scripts
- **Other:** Docker, Git/GitHub, Documentation & Runbooks

EDUCATION

- MSc in **Advance Computer Science with Data science (2:1)** **January 2023 – January 2024**
University of Strathclyde, School of Computing Science
- Bachelor of **Computer Applications (2:1)** **May 2016 – March 2020**
Savitribai Phule Pune university

PROJECTS

Real Time Stock Market Data Pipeline: ([GitHub Link](#))

- Designed & built a real-time pipeline (**Kafka, S3, Snowflake, dbt**) orchestrated with **Airflow**; **Docker Compose** stack; ~1-min latency to BI.
- Loaded **Snowflake** via stage-based PUT/COPY into RAW VARIANT; **idempotent micro-batches**, symbol-partitioned files, easy backfills.
- Modeled Medallion layers in dbt (Bronze/Silver/Gold) **with tests + lineage**; delivered Gold marts (latest KPIs, **OHLC, volatility**).
- Ensured reliability/observability: **Airflow retries/SLAs & logs**, durable raw zone, clear docs trusted near real-time reporting.

End-to-End Azure Data Engineering Lakehouse ([GitHub Link](#))

- Medallion (Bronze/Silver/Gold) on **ADLS Gen2**; parameterized, **metadata-driven ADF** with incremental loads, backfill controls, and JSON CDC watermarking.
- **Databricks Autoloader + Structured Streaming** refine Bronze to Silver **Delta**; **Unity Catalog** governance (metastore, external locations, access connector).
- **Delta Live Tables** for Gold: **SCD 2 dims** (sequence_by), SCD1 fact upserts, **Expectations + visual lineage DAG**; analytics-ready for **SQL Warehouse/Power BI**.
- Ops & DevEx: Python utilities (cleaning, de-dupe), checkpointing & idempotent runs, empty-file cleanup, **Logic Apps** alerts, secrets via **Managed Identity/Key Vault**, CI/CD with **Databricks Asset Bundles + GitHub**

DataBricks Declarative Pipelines ([GitHub Link](#))

- Built End-to-end **DLT Lakehouse** with **Medallion**; **unified batch + streaming** and automatic **DAG/lineage**.
- **Bronze:** streaming ingests with **Expectations** (warn/drop/fail) and Append Flow to merge multi-region sources; runtime audit metrics.
- Silver: **Auto-CDC** Type-1 upserts using reusable Python **utilities** (casting, de-dup, common transforms) for **incremental** accuracy.
- **Gold & Ops:** **SCD2** dimensions, Type-1 fact, and a **materialized revenue view**; **Unity Catalog** governance, parameterized dev/prod configs, **SQL seed scripts**, and **alerting**.

Azure Data Factory End-to-End (On-Prem + API + SQL): ([GitHub Link](#))

- ADF hybrid ingestion: **SHIR (on-prem shares)**, HTTP/REST, and Azure SQL **incremental loads** without watermark tables to **idempotent re-runs** and drift-tolerant landings.
- Orchestrated Bronze, Silver, Gold: **ADF Mapping Data** Flows standardize/clean/type-cast to **Silver Delta** with Alter Row upserts on business keys; **Gold** views (**joins, aggregations, dense-rank, Top-N**) optimized for BI.
- Centralized control via a **parent pipeline** chaining on-prem/API/SQL paths, **parameterizing file lists & batch sizes**, and parallelizing safe steps; less manual coordination and **predictable backfills/replays**.
- Shipped **ARM publish artifacts** and runbooks for **repeatable CI/CD**, with guardrails on **schema-drift vs validation**, **partitioning** options, and Gold snapshot overwrite strategy.

DBT Databricks Scd2-Data Quality: ([GitHub Link](#))

- Built a **Medallion** model with **dbt Core** on Databricks using env-driven **profiles.yml** and per-layer materializations.
- Enforced data quality with **dbt tests**: generic, **singular**, and **custom generic**.
- Implemented **SCD2** via **dbt snapshots** (timestamp strategy) for **time-travel analysis**.
- Drove reuse & speed with **Jinja macros**, **seeds**, **incremental** models, node selection, and **dbt build + docs/lineage** for CI.